

1. 解:

(1) 开环时

$$\Delta n_{op} = \frac{R I_d}{C_e}$$

$$C_e = \frac{U_N - I_N R}{n_N} = 0.13 \text{ Vmin/r}$$

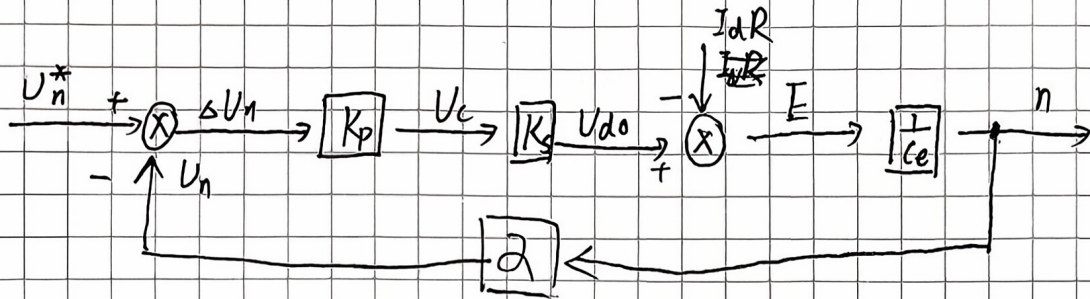
$$\Delta n_{op} = \frac{I_N R}{C_e} = \frac{2500}{13} \text{ r/min} \approx 192.3 \text{ r/min}$$

(2) 闭环时

$$D = \frac{n_N s}{\Delta n_{cl}(1-s)}$$

$$\Delta n_{cl} = \frac{n_N s}{D(1-s)} = \frac{75}{19} \text{ r/min} \approx 3.95 \text{ r/min}$$

(3)



(4)

$$n = \frac{K_p K_s U_n^*}{C_e (1+K)} - \frac{R I_d}{C_e (1+K)}, \quad K = \frac{K_p K_s \alpha}{C_e}$$

$$\text{当 } U_n^* = 15 \text{ V 时 } I_d = I_N = 12.5 \text{ A} \quad n = n_N = 1500 \text{ r/min}$$

$$\cancel{U_n = U_n^* = \Delta U_n} \quad \alpha = \frac{U_n}{n} \quad \text{静态时 } \Delta U_n \approx 0 \text{ 即 } U_n \approx U_n^*$$

闭环系统无静差近似, 则稳态时 $U_n \approx U_n^*$

$$\alpha = \frac{U_n^*}{n_N} = 0.01 \text{ Vmin/r}$$

(5)

$$K_p = \frac{K C_e}{K_s \alpha} = \frac{20 \times 0.13}{60}$$

$$\Delta n_{cl} = \frac{\Delta n_{op}}{1+K}$$

$$K = \frac{1861}{39}$$

$$K_p = \frac{K C_e}{K_s \alpha} = \frac{1861}{60} \approx 31$$